Dr. SHIVAM MAHAJAN

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PROFILE

I am a Computational Physicist with 7+ years of experience in developing high-performance computational models and simulations for complex materials. Proficient in Python and C++, I use data analysis to optimize material behavior. My expertise in molecular dynamics and statistical modeling is backed by a strong track record in interdisciplinary research and collaboration. I apply computational insights to tackle industry-relevant challenges, bridging theoretical research with practical applications.

PROFESSIONAL BACKGROUND

Research Fellow **May 2022–Present**
Nanyang Technological University, Singapore

- Conduct molecular dynamics simulations to study the structural, mechanical, vibrational, and dynamical evolution of complex amorphous systems.
- Write Python and C++ programs for statistical analysis and data processing.
- Develop and implement computational models to simulate material behavior at the atomic level and optimize simulation codes to enhance computational efficiency.
- Analyze large datasets to extract meaningful insights/patterns related to material properties.
- Prepare detailed technical reports and presentations to communicate research findings.

EDUCATION

1. **Doctor of Philosophy (PhD in Soft Condensed Matter Physics)** **July 2017–Mar 2022**
   Nanyang Technological University, Singapore
   - Cumulative Grade Point Average (CGPA): 4.88/5
   - Conducted molecular dynamics simulations of amorphous systems to develop statistical models for multi-agent interactions and improve material behavior understanding.
   - Analyzed correlations in crowded networks and force transmission in materials to gain insights into their mechanical properties.
   - Collaborated with experimental researchers to validate simulations, ensuring consistency with real-world data and advancing material characterization.
2. **Post Graduate study (Master of Science in Physics)** **July 2014-June 2016**
   Guru Nanak Dev University, Amritsar, India
   - Cumulative Grade Point Average (CGPA): 9.56/10
   - Deposited thin films of organo-metallic compounds and examined their structural and optical properties using various techniques.
3. **Undergraduate study: (Bachelor of Science)** **July 2011-June 2014**
   University of Jammu, India
   - Percentage: 84.9%

KEY SKILLS

- Programming Languages: Python, C++.
- Simulations: Molecular dynamics simulations (LAMMPS), Monte-Carlo simulations, Parallel programming.
- Analytics: Computational Methods, Statistics, Machine Learning.
- Documentation: LaTeX, MS-Office.
- Languages: English, Hindi, Punjabi, Dogri.

SUMMER INTERNSHIPS

1. Summer Research Fellow **May 2015-July 2015**
Bhabha Atomic Research Centre (BARC), Mumbai
Selected by **Indian Academy of Sciences (IAS), Banglore**

- Studied the interaction of silica nanoparticles with proteins.
- Techniques: Small Angle Neutron Scattering (SANS), Dynamic Light Scattering (DLS).

2. Summer Research Fellow **June 2015-July 2013**
Indian Institute of Science Education and Research (IISER), Mohali
Selected by **Indian Academy of Sciences (IAS), Banglore**

- Project on Basic Laboratory Techniques and Stereoselective Synthesis of Oxindoles
- Techniques: Column Chromatography, TLC Formation.

SELECTED PUBLICATIONS

full list on Google Scholar

1. Revealing the Geometrical and Vibrational Properties of Defects in Amorphous Solids – ***Physical Review E, 2025***
2. Heterogeneous attenuation of sound waves in three-dimensional amorphous solids – ***Physical Review E, 2024***.
3. Quasi-localized vibrational modes, boson peak and sound attenuation in model mass-spring networks - ***SciPost Physics, 2023***
4. Unifying the vibrational properties in amorphous solids - ***Physical Review Letters, 2021***
5. Emergence of linear isotropic elasticity in amorphous and polycrystalline materials - ***Physical Review E, 2021***
6. Designing phononic band gaps with sticky potentials - ***Frontiers in Physics, 2021***
7. Reverse Janssen Effect in narrow granular columns - ***Physical Review Letters, 2020***

AWARDS AND ACHIEVEMENTS

1. **Gold Medal** for the best PhD Thesis in Physics, Materials Research Society of Singapore, 2022.
2. **3-Minute Thesis Competition**: Department Winner, University 2nd Runner-up, National Participant, 2020.
3. **Teaching Assistant Certification** from the Teaching, Learning, and Pedagogy Department, NTU, 2019.
4. **Lectureship Eligibility** for Indian Universities, CSIR-NET (All India Rank: 22), 2017.
5. **GATE**: Top 5% in All India Graduate Aptitude Test for Engineering in Physics, 2016-2017.
6. **Secured** full scholarship for my doctoral studies at NTU, 2017.
7. **MSc Hons (Physics)**: Ranked 3rd in University, 2016.
8. **BSc**: Ranked 1st in College and 3rd in University, 2014.
9. **Summer Research Internships** at BARC and IISER, selected twice by Indian Academy of Sciences, 2013, 2015.
10. **Higher Secondary Board**: Ranked 1st in District, 2011.

CONFERENCES

- **Invited Talk**: "Dynamics of Supercooled Liquids in the Energy Landscape" at *Seeking Unifying Principles in Equilibrium and Driven Materials*, Italy (2024).
- **Contributed Talk**: "Reverse Janssen Effect in Narrow Granular Columns" at *Soft Matter, Fluids, Interfaces*, Vietnam (2024).

- **Contributed Talk**: "Mechanism of Phonon Attenuation in 3D Amorphous Solids" at the *Institute of Physics*, Singapore (2023).
- **Contributed Talk**: "Mechanism of Phonon Attenuation in 3D Amorphous Solids" at the *9th International Discussion Meeting on Relaxations in Complex Systems*, Japan (2023).
- **Contributed Talk**: "Unifying the Vibrational Anomalies of Amorphous Materials" at the *APS March Meeting* (2022).
- **Poster Presentation**: "Reverse Janssen Effect in Narrow Granular Columns" at the *APS March Meeting* (2022).
- **Contributed Talk**: "Elasticity and Ordering in the Structure of Solids at Different Length Scales" at the *First Rencontres du Vietnam, Soft Matter Science* (2019).
- **Poster Presentation**: "Anomalous Janssen Effect in Narrow Granular Columns" at *Designer Soft Matter*, Singapore (2018).

LEADERSHIP

1. **Advisor** in **NTU World of Wisdom (WoW)** team (2022-2023): Guiding the student leaders in organization of various events.
2. **Team lead** of Operations & Logistics team of **TEDxNTU**, the biggest TED conference in Singapore (2020-22): Led the team of about fifty people and contributed towards the successful organization of the conference.
3. **Director (Emeritus)** for Graduate Students' club of the School of Physical and Mathematical Sciences (2020-22): Organized monthly seminars and Nobel Laureate Public Lectures with financial support from the Institute of Advanced Studies.
4. **President** for Graduate Students' club of the School of Physical and Mathematical Sciences (2019-20): Organized various social, recreational and academic events.

HOBBIES

Travelling, Reading novels, Playing cricket, table-tennis, sudoku